

Raven Cruz

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SUMMARY

Designer and developer with a B.S. in Computer Science (Game Design) from UC Santa Cruz, focused on building tools that make creative work faster and more approachable for non-technical collaborators. Five plus years in the Adobe suite (Photoshop and Lightroom, with working knowledge of Premiere Pro) alongside a programming background centered on JavaScript and TypeScript. Two years of research building AI-driven pipelines and evaluating model outputs, with two peer-reviewed publications.

SKILLS

Design & Creative Tools: Adobe Photoshop, Lightroom, Premiere Pro; game engines (various) and design

Languages: JavaScript, TypeScript, HTML5, C++, C#, Lua

Technical: AI/LLM data pipelines, data-driven design, rapid prototyping

EXPERIENCE

Design Lead & Narrative Lead | AstroFrame (UC Santa Cruz Senior Capstone) | Steam Early Access, Summer 2026 | 2026

- Led design on a 2D roguelike, architecting a spreadsheet-driven JSON data pipeline that let the team manage game state, balance, and dialogue without touching engine code.
- Wrote the world's history, factions, and characters as narrative lead, and designed the randomization systems behind the game's replayability.

Undergraduate Researcher | UC Santa Cruz | 2024 – 2026

- *Published:* [Conversational Interactions with Procedural Generators using Large Language Models](#), FDG 2025.
 - Built a content generator with an annotation system that automatically described generated content in natural language, producing structured training data for LLMs. Evaluated model-generated responses for accuracy. Developed a critical, practical eye for where AI helps a creative workflow and where it needs a human check.
- *Published:* [SketchTiler: Co-Creative Sketch-to-Tilemap Conversion Using Wave Function Collapse](#), IEEE CoG 2025.
 - Built a sketch-to-tilemap tool that turns rough user sketches into finished, fully tiled levels, lowering the barrier for people who are not fluent in tile-based design; profiled and redesigned the underlying Wave Function Collapse implementation with a hierarchical approach, cutting average runtime from roughly 10 seconds to 1 second (10x improvement).

Course Tutor | UC Santa Cruz | 2025 – 2026

- Provided written and verbal technical feedback across 5 CS: Game Design courses built primarily on JavaScript/HTML5, including the Phaser framework and p5.js. Adapted communication style to students at varying skill levels.
- Diagnosed errors and edge cases across a wide range of unfamiliar codebases and tools; translated findings into clear, actionable explanations. Participated in informal code review and pair-debugging sessions.

Assistant Manager | Crossroads Trading Co. | July 2017 – January 2021

- Led cross-functional teams, authored operational documentation, and trained staff at multiple levels in a fast-paced, independently managed environment.

EDUCATION

B.S. Computer Science: Game Design | UC Santa Cruz | Dean's Honors | March 2026

A.S. Computer Science & A.S. Mathematics | American River College | Highest Honors | May 2023